

Project Proposal

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SI 305: Introduction to Information Analysis

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March 22, 2026

1. Problem Scoping:

1.1. Identify the Problem:

Children in Lansing living in low-income, pre-1978 rental housing remain at higher risk of elevated blood lead levels (EBLLs), despite statewide declines in lead exposure¹.

Local lead abatement initiatives like Lead Safe Lansing (LSL) operate under tight budgets but lack systematic data on the socioeconomic status and housing precarity of the households they serve, making it difficult to ensure that limited resources reach the most vulnerable renters between 2011 and 2022.

1.2. Problem Statement:

In 2024, 140,498 children under age 6 in Michigan were tested for EBLLs, and 5,846 (4.2%) had blood lead levels at or above 3.5 µg/dL², indicating that lead exposure remains a serious statewide public health concern. Elevated blood lead levels are associated with cognitive deficits, behavioral problems, and cardiovascular harms such as increased blood pressure, underscoring the importance of sustained prevention and remediation efforts.

In Lansing, children in low-income households are disproportionately exposed because they are concentrated in older, pre-1978 rental properties in historically underinvested neighborhoods where deteriorating lead-based paint and contaminated dust are primary exposure pathways³. Guest lectures from LSL staff Erin Buitendorp and Tyler Blakey emphasize that these risks fall especially heavily on children under 6 and pregnant women, highlighting the need to prioritize high-risk households in local abatement work⁴.

¹ Michigan Department of Health and Human Services, “CHILDHOOD LEAD POISONING PREVENTION IN MICHIGAN” (Michigan Department of Health and Human Services, April 2025), 1, accessed March 27, 2026, https://www.michigan.gov/mileadsafe/-/media/Project/Websites/mileadsafe/Reports/FY-2024-CLPPP-LEGISLATIVE-REPORT_April-2025.pdf.

² Lanphear, Bruce, Ana Navas-Acien, and David C. Bellinger. “Lead Poisoning.” *New England Journal of Medicine* 391, no. 17 (2024): 1621–31. <https://doi.org/10.1056/NEJMr2402527>.

³ “Child Lead Exposure Elimination Commission,” n.d., <https://www.michigan.gov/mdhhs/doing-business/commissions-boards/childhoodlead/child-lead-exposure-elimination-commission>.

⁴ Erin Buitendorp, “L05: Erin Buitendorp Visit” (Lansing, United States of America, Jan the Lansing area data the nuary 26, 2025) signifies

At the same time, LSL’s administrative records focus on properties and projects rather than residents, and they do not systematically record tenant income, housing stability, or other indicators of socioeconomic vulnerability. This structural data gap makes it difficult for city officials, program staff, and state agencies to evaluate whether scarce abatement resources are reaching low-income renters in older housing who are least able to self-finance remediation and most vulnerable to displacement, or whether more advantaged property owners are disproportionately benefiting from the program

2. Problem Formulation:

2.1. Developing Research Questions:

RQ1: How does the concentration of completed lead abatement projects across Lansing's ZIP codes from 2011 to 2022 align with ZIP-level income and blood lead level risk among young children?

RQ2: Among completed Lead Safe Lansing abatement projects completed between 2011 and 2022, are higher rehabilitation costs associated with the inflation rate or higher elevated blood lead levels?

2.2. Understanding the Motivation

2.2.1. RQ1:

RQ1 is motivated by equity and accountability concerns arising from the absence of resident-level socioeconomic data in LSL’s internal records. By linking project locations to neighborhood-level indicators such as income and poverty, this question focuses on who benefits from completed abatement projects and whether they are concentrated in neighborhoods with higher EBLL prevalence among children.

The results will be relevant to LSL program staff, city housing officials, and the Michigan Department of Health and Human Services (MDHHS), who are responsible for ensuring that lead hazard control efforts prioritize at-risk families. While the analysis cannot show whether the program is “effectively reaching” all eligible low-income households, it can reveal whether completed projects are at

least aligned spatially with lower-income, higher-risk neighborhoods where young children live.

2.2.2. **RQ2:**

This question is motivated by a core limitation in interpreting rehabilitation costs across LSL projects: cost variation may reflect household and property differences, but it may equally reflect broader economic forces outside the program's control, such as inflation, labor market conditions, and shifts in contractor availability. Without accounting for these macroeconomic factors, it is easy to misattribute higher costs in certain years or among certain groups to programmatic decisions when they actually reflect economy-wide trends.

LSL program staff, city housing officials, and MDHHS would benefit from understanding which cost drivers are within the program's influence and which are external. For example, if rehabilitation costs rise sharply for lower-income households even after controlling for inflation and unemployment, that would suggest structural inequities in how resources are deployed, a signal to revisit program design. Conversely, if macro conditions explain most of the cost variation, that would validate current practices while highlighting the need for inflation-adjusted budgeting in future grant cycles.

This question is explicitly descriptive and predictive, not causal. The regression results will identify associations between cost and these variables, but will not establish that any single factor directly caused higher rehabilitation costs.

2.3. **Deriving Plans for Research Questions**

Internal Datasets:

- [ProjectMasterExport_4.24.25 Raw Data_Single Family and Rental Abatement.xlsx](#)

I will focus on the “STREET” variables to determine which ZIP code it belongs to, so I can later assign it a ZIP code using external research. I

will also pay attention to the “REHABTOTAL,” as I plan to measure the average costs for each ZIP code in this dataset. The “COMMENTS” variable will help me to filter out which project is dropped from the program. By using this dataset and its listed variables, I will be able to answer both RQs. For the first one, by understanding the location, I can later aggregate the ZIP code to see which area has more projects than the other. By understanding the project's cost, I can gain insight into where LSL distributes its grant and the amount associated with it. This part touches on RQ1 by discussing how it addresses at-risk neighborhoods. It also links to RQ2 because I want to investigate how yearly costs correlate with changes in funding across cycles.

- [data_152512.csv](#)

I will focus on the “Geographic Area(s)”, “Blood Lead Reference Value”, and “Percent of Children Tested with an EBLL.” The “Geographic Area(s)” will allow me to match the ZIP codes with the LSL’s and the NaNDA’s data, as mentioned in the External Datasets section. This will provide a more holistic view of which areas are at risk and how LSL has worked with those areas, which is a core part of RQ1. The “Blood Lead Reference Value” will act as a filter. I will choose 3.5 because it matches the CDC's value⁵. Last but not least, “Percent of Children Tested with an EBLL” will be critical when answering both questions. In RQ1, my goal is to check how this value decreases throughout the scope (2011-2024) among different ZIP codes. In RQ2, I will examine if an increase in government funding will lead to more homes being abated, and lower the percentage of children with elevated blood lead levels.

External Datasets:

- [National Neighborhood Data Archive \(NaNDA\): Socioeconomic Status and Demographic Characteristics of Census Tracts and ZIP Code Tabulation Areas, United States, 1990-2022](#)

⁵ “CDC Updates Blood Lead Reference Value,” Childhood Lead Poisoning Prevention, April 2, 2024, <https://www.cdc.gov/lead-prevention/php/news-features/updates-blood-lead-reference-value.html>.

Dataset description: It contains neighborhood-level socio-demographic indicators (e.g., income, poverty, housing, race/ethnicity, age structure) for U.S. census geographies, typically at tract and ZIP code levels, constructed to support neighborhood health and inequality research. The dataset was collected and curated by the National Neighborhood Data Archive (NaNDA) team at the University of Michigan’s Institute for Social Research. A detailed codebook and metadata are available on the study page under “Data Review,” describing variables, construction methods, and geographic units. The main limitation of the data is that it is only available until 2022, which does not cover our scope. I have found a potential dataset that can fill the current gap, such as the American Community Survey 1-year data between 2023 and 2024.

Value beyond your current inventory: It provides information on the socioeconomic status of many locations across the United States, helping fill gaps in the data from LSL or Professor Hoover.

- [Lead Safe Lansing 2, 3 & 4 Grants](#)

Data Description: It contains the dollar value of the grant awarded to LSL to carry out its projects, with the start and end dates of the projects. A methodology explaining how USAspending collects this data is available under “Data sources and methodology” on every page. The core limitation of this dataset is that I do not have access to the number of projects completed during these periods, unlike the LSL’s final report after each period. Because of this, I will conduct a separate analysis to match each period with the number of projects completed during that period.

Value beyond your current inventory: The other two datasets do not contain any information on funding, specifically the dollar value LSL receives from the government. Thus, this dataset will enable us to assess the efficiency of LSL in allocating funding to complete projects.

2.3.1. **RQ1:**

To examine whether completed abatement projects align with lower-income, higher-risk areas, I will integrate three datasets: LSL's abated project data, MiTracking's blood lead data, and neighborhood socioeconomic indicators from the National Neighborhood Data Archive (NaNDA), supplemented by recent American Community Survey data where needed. After cleaning the LSL data to retain only completed projects and extracting standardized property locations (e.g., ZIP code), I will join MiTracking and NaNDA by ZIP code to obtain EBLR rates among tested children under 6 and measures such as the share of households under 40,000 dollars and between 40,000 and 79,999 dollars.

Analytically, I will:

- Compute the count and rate of completed LSL projects per ZIP code.
- Compare average EBLR rates and income measures between ZIP codes with higher versus lower densities of completed projects using descriptive statistics and simple group comparisons (e.g., means, boxplots, and non-parametric tests if appropriate).
- Use matrix-style visualizations such as correlation heatmaps (e.g., showing relationships among project density, EBLR rates, and income measures) and clustered bar charts to summarize how project activity varies across income and risk categories, without relying on geographic maps.

These steps will be implemented in Python using pandas for data wrangling and seaborn/matplotlib for visualization, building on my prior coursework and data analysis experience

2.3.2. RQ2:

I will construct grant-cycle-level and project-level measures of outputs and costs using LSL's project dataset and external grant data for LSL1–LSL4 between 2006 and 2025. First, I will assign each project to a grant cycle using FY_COMPLT, then aggregate within each cycle to obtain total completed units and average cost per completed unit.

Recognizing that there are only three grant cycles, I will supplement this with project-level analyses, computing rehabilitation costs per unit and examining how they vary by year, income tier, building age, and race across approximately 416 completed abatement records. I will merge annual macroeconomic indicators, such as the Michigan unemployment rate, a construction cost index, and HUD Fair Market Rents, by fiscal year, then run a linear regression with REHABTOTAL as the outcome. Differences across years and cycles may reflect inflation, project complexity, and labor market conditions, and all findings will be interpreted accordingly.